1 Important Constants & Equations

This week you covered the basis of the study of electricity, the electric charge, electric force, and the electric field.

For two charged particles, q_1 and q_2, separated by a distance r, the force felt by *both* particles is:

$$F = K\frac{|q_1|\,|q_2|}{r^2} \qquad \text{(Coulomb's Law)}$$

Electric forces can simply be added (i.e. they follow the principle of superposition):

$$\vec{F}_{net} = \vec{F}_{1\,\mathrm{on}\,i} + \vec{F}_{2\,\mathrm{on}\,i} + \cdots + \vec{F}_{n\,\mathrm{on}\,i} \qquad \text{(Principle of Superposition for Forces)}$$

The force felt by a charge q_1 from a charge q_2 separated by a distance r is proportional to the electric field generated by q_2:

$$F \propto E \implies F_{q_2\,\mathrm{on}\,q_1} = Eq_1 \implies K\frac{|q_1|\,|q_2|}{r^2} = Eq_1 \implies E = K\frac{|q_1|}{r^2}$$

$$E = K\frac{|q_1|}{r^2} \qquad \text{(Electric Field of a Point Charge)}$$

The following constants are important for the study of electricity:

$e = 1.60 \times 10^{-19}$ C (Fundamental Unit of Charge)

$m_p = 1.67 \times 10^{-27}$ kg (Mass of Proton)

$m_e = 9.11 \times 10^{-31}$ kg (Mass of Electron)

$K = 8.99 \times 10^{9}$ Nm^2/C^2 (Electrostatic Constant)

$\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 (Permittivity Constant)

2 Problems

1. An object has a total $\pm 1C$ (coulomb) magnitude of charge.

   (a) Assume the charge is $-1C$ and consists only of electrons, how many electrons does the object have?

   (b) Using your answer from part (a), what is the mass of this object?

   (c) Now, if the charge is $+1C$ and consists only of protons, how many protons does the object have?

   (d) Using your answer from part (d), what is the mass of this object?

(e) What is the mass-ratio between the proton and electron?

$$\frac{m_p}{m_e} =$$

2. Coulomb's Law is written as:

$$F = K\frac{|q_1|\,|q_2|}{r^2}.$$

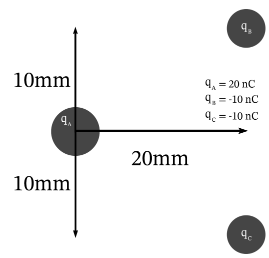
We also know that the permittivity constant $\epsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$, is defined as:

$$\epsilon_0 = \frac{1}{4\pi K}.$$

(a) Rewrite Coulomb's law using ϵ_0.

(b) Two charged objects, $q_1 = 5nC$ and $q_2 = -7nC$, are separated by a distance $d = 1mm$. Find the magnitude of force felt by *both* charged objects.

3. Three charged particles are interacting as shown in the diagram below.



(a) Find a position vector to describe the distance between q_A and q_B. Then determine it's magnitude.

$$\vec{r}_{AB} = \underline{\hspace{2cm}}\,\hat{\imath} + \underline{\hspace{2cm}}\,\hat{\jmath} \quad (m)$$

$$|\vec{r}_{AB}| = \underline{\hspace{2cm}}\ m$$

(b) Find a position vector to describe the distance between q_A and q_C. Then determine it's magnitude.

$$\vec{r}_{AC} = \underline{\hspace{2cm}}\,\hat{\imath} + \underline{\hspace{2cm}}\,\hat{\jmath}$$

$$|\vec{r}_{AC}| = \underline{\hspace{2cm}}\ m$$

(c) What do you notice about the distances in the y-direction? What significance does this hold for our problem?

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(d) Express the force $\vec{F}$ felt by q_A in vector notation. What is the magnitude of this force?

4. A charged particle q_1 with an unknown magnitude of charge pulls on another charged particle q_2 with a known charge of $+5\ nC$.

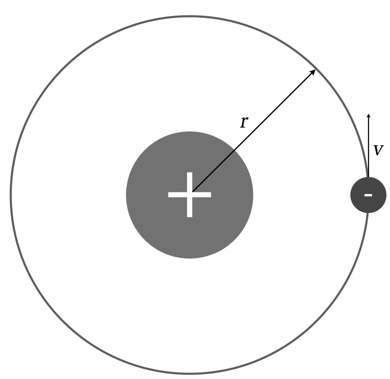
   (a) Considering that q_1 *pulls* q_2, does q_1 have a positive or negative charge? (Circle one)

   (b) The distance between q_1 and q_2 is $5\ \mu m$. Diagram the situation below.

   (c) What is the magnitude of the electric field generated by q_2 in the place where q_1 resides?

   (d) Recall that the force acting on both particles must be equal in magnitude. Using this, find the charge of q_1 if the force felt by **both** particles has a magnitude of $5\ N$.

5. **(CHALLENGE PROBLEM).** If you have taken an introductory chemistry course, you may have learned about the "Bohr Model of the Atom," a model that describes electrons rotating about the proton in perfect circles (somewhat similar to how planets orbit about the Sun). While there are issues with the model, it is a useful introduction. For this problem, assume that the model is accurate.



As shown in the model above, an electron rotates about a proton with a velocity v at a radius r within a Hydrogen atom. Ignoring any effect of gravity:

(a) Find an expression for a radius r which the electron *must* rotate at.

(b) Let the velocity be $v = 5 \times 10^6\ m/s$.

i. Find the radius r.

ii. Find the kinetic energy K of the electron.

iii. **(BONUS).** What would happen if the electron were to lose kinetic energy over time?